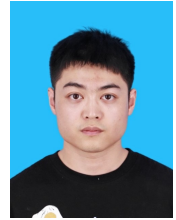


Pengfei Ma

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Research Interests

- My research centers on computational cardiology, with a primary focus on fully GPU-accelerated fluid–structure interaction for whole-heart simulations. Building on this foundation, I extend my work to peripheral vascular circulation, electromechanical coupling of cardiac tissue, and coronary perfusion modeling. Ultimately, I aim to translate these high-fidelity, high-performance simulations into clinically relevant tools that support diagnosis, treatment planning, and personalized medicine.

Education

- 2019 – 2025 **Ph.D.** in Mathematics (successfully defended in December 2025).
School of Mathematics and Statistics, Northwestern Polytechnical University
- 2018 – 2019 **M.Sc.** in Computational Mathematics (Transferred to Ph.D. Program)
School of Mathematics and Statistics, Northwestern Polytechnical University
- 2014 – 2018 **B.Sc.** in Information and Computing Science
School of Science, Northwestern Polytechnical University

Research Publications

Journal Articles

- 1 **P. Ma**, L. Cai, X. Wang, and H. Gao, “AFSI: Automated fluid-structure interaction solver development for nonlinear solid mechanics,” *arXiv preprint arXiv:2509.00014*, 2025.
- 2 **P. Ma**, L. Cai, X. Wang, and H. Gao, “Fully GPU-accelerated, matrix-free immersed boundary method for complex fiber-reinforced hyperelastic cardiac models,” *Computer Methods in Applied Mechanics and Engineering*, vol. 447, p. 118 353, 2025, ISSN: 0045-7825.
- 3 X. Wang, L. Cai, **P. Ma**, and H. Gao, “An efficient staggered scheme for solving the poromechanics problem of quasi-static cardiac perfusion,” *International Journal for Numerical Methods in Biomedical Engineering*, vol. 41, no. 4, e70030, 2025.
- 4 Q. Chen, L. Cai, F. Jing, **P. Ma**, X. Luo, and H. Gao, “On the immersed boundary method with time-filter-sav for solving fluid–structure interaction problem,” *Journal of Scientific Computing*, vol. 100, no. 2, p. 45, 2024.
- 5 Y. Liu, L. Cai, Y. Chen, **P. Ma**, and Q. Zhong, “Variable separated physics-informed neural networks based on adaptive weighted loss functions for blood flow model,” *Computers & Mathematics with Applications*, vol. 153, pp. 108–122, 2024.
- 6 **P. Ma**, L. Cai, X. Wang, Y. Wang, X. Luo, and H. Gao, “An unconditionally stable scheme for the immersed boundary method with application in cardiac mechanics,” *Physics of Fluids*, vol. 36, no. 8, 2024.
- 7 L. Cai, Y. Hao, **P. Ma**, G. Zhu, X. Luo, and H. Gao, “Fluid-structure interaction simulation of calcified aortic valve stenosis,” *Mathematical Biosciences and Engineering*, vol. 19, no. 12, pp. 13 172–13 192, 2022.
- 8 L. Cai, J. Jiao, **P. Ma**, W. Xie, and Y. Wang, “Estimation of left ventricular parameters based on deep learning method,” *Mathematical Biosciences and Engineering*, vol. 19, no. 7, pp. 6638–6658, 2022.

Skills

Coding	■ C++, Python, CUDA, MPI
Numerical	■ Finite Element Method, Finite Difference Method, Immersed Boundary Method

Miscellaneous Experience

Participated Projects

2019.1–2022.12	■ National Natural Science Foundation of China General Program (11871399), Efficient Numerical Methods and Applications for Electrophysiological Coupling in Injured Hearts
2023.1–2026.12	■ National Natural Science Foundation of China General Program (12271440), Multiscale and Multiphysics Coupled Modeling and Computation of Myocardial Perfusion

Software Development

AFSI	■ An open-source Python package for automated fluid-structure interaction modeling. https://github.com/npuheart/afsi
NPUHEART	■ A high-performance GPU-based framework for large-scale fluid-structure interaction simulations. (Not yet open-sourced)

References

Supervisors	■ https://www.maths.gla.ac.uk/~xl/team.html https://teacher.nwpu.edu.cn/caili.html
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